SLIDE-QUIZ EXERCISE · EVOLLAND01

Evolution of Terrestrial Life

The Colonization of Land Over 500 Million Years

TEACHER GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EXERCISE OVERVIEW

Evolution of Terrestrial Life

This exercise follows life's conquest of Earth's land surface across 500 million years, from microscopic crusts on bare rock to forests, dinosaurs, flowering meadows, and finally us. Eight pages walk forward through geologic time, one period at a time; each shows what the plants and animals were like, what the atmosphere was doing, and what changed. Two closing pages step back to explain why it all happened: the slow buildup of oxygen and the ozone shield, the handful of evolutionary innovations that freed life from the water, and the volcanoes, drifting continents, and asteroid impacts that repeatedly reset the stage. Every content page has a voiceover you can play, images you can hover over to animate, and one graded question, ten in all.

The Exercise, Page by Page

Page	What you'll do
Introduction	A short video explains what to expect on the pages that follow.
1. The Pioneers	Ordovician–Silurian (500–420 Ma): the first life on land. One question.
2. The First Forests	Devonian (420–360 Ma): the first trees and the first four-legged animals. One question.
3. The Coal Forests	Carboniferous (360–299 Ma): giant plants, giant insects, and the oxygen peak. One question.
4. Supercontinent Pangaea	Permian (299–252 Ma): seed plants, synapsids, and the worst mass extinction. One question.
5. Recovery & Revolution	Triassic (252–201 Ma): the first dinosaurs and the first mammals. One question.
6. Age of Giant Dinosaurs	Jurassic (201–145 Ma): sauropods and the first birds. One question.
7. Flowering Plant Revolution	Cretaceous (145–66 Ma): flowers, pollinators, and the end of the dinosaurs. One question.
8. Age of Mammals & Humans	Cenozoic (66 Ma–present): grasslands, mammals, and Homo sapiens. One question.
9. The Atmospheric Foundation	How oxygen and the ozone layer made life on land possible. One question.
10. The Big Picture	The key innovations and the geologic forces behind them. One question.
Conclusion	A closing video, your score, and the Submit Quiz button.

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Learning Objectives

After this exercise you should be able to:

- Explain why oxygen and the ozone layer had to come first before complex life could live on land.
- Put the major steps of plant evolution in order: crusts, vascular plants, trees, seed plants, and flowering plants.
- Describe how high Carboniferous oxygen made giant insects and other arthropods possible.
- Explain how the amniotic egg freed vertebrates from returning to water to reproduce.
- Identify the causes and the consequences of the End-Permian and End-Cretaceous mass extinctions.
- Describe the rise of dinosaurs, birds, and mammals through the Triassic, Jurassic, and Cretaceous.
- Explain the co-evolution of flowering plants and their insect pollinators.
- State when and where Homo sapiens evolved, and place it on the 500-million-year timeline.

QUICK REFERENCE

Key Terms

Term	Definition
Ma / Ga	Millions of years ago / billions of years ago. 430 Ma means 430 million years ago.
Cyanobacteria	Photosynthesizing microbes. They put the first oxygen into Earth's atmosphere and formed some of the earliest crusts on land.
Lichen	A partnership of a fungus with algae or cyanobacteria. Lichens grow on bare rock and slowly break it down into soil.
Bryophytes	Simple, low-growing plants without vascular tissue, such as mosses.
Mycorrhizae	Partnerships between fungi and plant roots; the fungus helps the plant take up water and nutrients.
Vascular tissue	A plant's internal plumbing: xylem carries water up, and phloem carries food around. It lets plants grow tall.
Tetrapod	A vertebrate with four limbs: amphibians, reptiles, birds, and mammals, and their ancestors.
Amniotic egg	A self-contained egg with its own water supply and a protective shell, so it can be laid on dry land.
Lycophytes	Club mosses. Today they are small, but in the Carboniferous some, such as Lepidodendron, were trees 50 m tall.
Gymnosperms	Seed plants whose seeds are not enclosed in fruit: conifers, cycads, and ginkgos.
Angiosperms	Flowering plants. Their seeds develop inside fruit.

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Term	Definition
Synapsids	The group of animals that includes mammals and their ancestors, such as the Permian gorgonopsids.
Archosaurs	The group that includes dinosaurs, pterosaurs, crocodilians, and birds.
Sauropods / Theropods	Sauropods: giant, long-necked, plant-eating dinosaurs. Theropods: two-legged, mostly meat-eating dinosaurs, the group that gave rise to birds.
Hominins	Humans and our extinct relatives after our lineage split from the other apes.
Pangaea	The supercontinent that held nearly all the land on Earth in the Permian and Triassic.
Mass extinction	A geologically brief event in which a large share of all species dies out.
Co-evolution	Two groups evolving together, each in response to the other, like flowers and their pollinators.
Great Oxidation Event	The time, about 2.4 Ga, when oxygen first began to build up in the atmosphere.
Ozone layer	A layer of ozone (a form of oxygen) high in the atmosphere that blocks the Sun's harmful ultraviolet (UV) light.

QUICK REFERENCE

The Timeline

Period	When	Highlights
Ordovician–Silurian	500–420 Ma	Microbial and lichen crusts; mosses; Cooksonia, the first vascular plant; millipedes ashore
Devonian	420–360 Ma	First forests (Archaeopteris); deep roots make true soil; Tiktaalik; first tetrapods
Carboniferous	360–299 Ma	Coal swamps; oxygen peaks near 35%; giant insects; amniotic egg; first reptiles
Permian	299–252 Ma	Pangaea; gymnosperms and synapsids dominate; End-Permian extinction (252 Ma)
Triassic	252–201 Ma	Recovery; first dinosaurs (~230 Ma), pterosaurs, and mammals (~225 Ma)
Jurassic	201–145 Ma	Giant sauropods; conifer forests; Archaeopteryx (~150 Ma)
Cretaceous	145–66 Ma	Flowering plants and pollinators; T. rex; End-Cretaceous extinction (66 Ma)
Cenozoic	66 Ma–present	Mammals diversify; grasslands spread; Homo sapiens (~300,000 years ago)

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QUICK REFERENCE

Numbers to Know

Value	What it refers to
500 million years	Span of the exercise, from the first crusts on land to today
2.4 Ga	Great Oxidation Event: oxygen begins to build up in the atmosphere
~430 Ma	Cooksonia, the first vascular plant, 2–6 cm tall
10–15%	Atmospheric oxygen when life first reached land
35%	Peak oxygen in the Carboniferous
21%	Oxygen in the atmosphere today
70 cm / 2.5 m	Wingspan of the dragonfly Meganeura / length of the millipede Arthropleura
50 m	Height of the Carboniferous scale tree Lepidodendron
~325 Ma	The amniotic egg appears
252 Ma, 96%	End-Permian extinction and the share of species it killed
66 Ma, 75%	End-Cretaceous extinction and the share of species it killed
10 km	Width of the Chicxulub asteroid
~300,000 years	Age of our species, Homo sapiens, which evolved in Africa

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EXERCISE CONTENT

Page-by-Page Summary

Part 1 — The Pioneers

Ordovician–Silurian, 500–420 Ma. The first life on land was microscopic. **Cyanobacteria** and **lichens** formed thin crusts on bare rock, and lichens began weathering that rock into the first soil. Simple mosses, or **bryophytes**, followed around 470 Ma, and by about 450 Ma fungi were forming partnerships with plant roots (**mycorrhizae**).

Around 430 Ma came **Cooksonia**, only 2–6 cm tall but the first plant with **vascular tissue**: internal plumbing to carry water. By about 428 Ma **millipedes** had ventured ashore, feeding on decaying plant material and needing moist surroundings to survive.

Oxygen was only about 10–15% of the atmosphere, compared with 21% today, but the ozone layer was strong enough to make the surface livable, if still harsh.

Part 2 — The First Forests

Devonian, 420–360 Ma. Plants grew taller. Club mosses (**lycophytes**) reached a meter, true ferns evolved complex leaves, and around 385 Ma the first forests appeared. Their best-known tree, **Archaeopteris**, grew up to 30 m tall.

Archaeopteris's great innovation was underground: roots more than a meter deep. They broke up rock and created **true soil**, and the faster weathering of rock changed the chemistry of the atmosphere and oceans.

In the water, **Tiktaalik** (375 Ma) was a “fishapod”: a fish with sturdy, limb-like fins, transitional between fish and four-limbed animals. By 365 Ma the first **tetrapods**, such as Acanthostega and Ichthyostega, had appeared. Spiders hunted in the undergrowth.

Part 3 — The Coal Forests

Carboniferous, 360–299 Ma. This was the age of giants. Scale trees (**Lepidodendron**) grew 50 m tall, tree ferns 20 m, and giant horsetails (Calamites) 18 m. These vast swamp forests were so productive that their buried, undecayed remains became today's **coal**.

Atmospheric **oxygen peaked at about 35%**, compared with 21% today. Insects and other arthropods take in oxygen by diffusion through their bodies, which normally limits how big they can grow; extra oxygen raised that limit. The dragonfly **Meganeura** had a 70 cm wingspan, and the millipede **Arthropleura** reached 2.5 m.

Around 325 Ma came the **amniotic egg**, an egg with its own water supply and a shell, which could be laid on dry land. It freed vertebrates from returning to water to reproduce, and the first reptiles and synapsids followed.

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Part 4 — Supercontinent Pangaea

Permian, 299–252 Ma. The continents had gathered into the supercontinent **Pangaea**, and its interior grew dry. **Gymnosperms** (conifers, cycads, ginkgos) and seed ferns such as Glossopteris took over, while spore plants like lycophytes and horsetails declined. Seeds did better than spores in dry environments.

Synapsids, the group that includes mammals' ancestors, ruled the land. **Gorgonopsids** were saber-toothed predators up to 3.5 m long, and large, barrel-chested plant-eaters roamed. The ancestors of dinosaurs and the first beetles appeared.

The period ended with the **End-Permian extinction** (252 Ma), the worst in Earth's history. Massive volcanic eruptions in the **Siberian Traps** released carbon dioxide, causing global warming, ocean acidification, and oxygen-starved (anoxic) oceans. About 96% of species died out.

Part 5 — Recovery & Revolution

Triassic, 252–201 Ma. Life recovered, but it looked very different. Gymnosperms still dominated the plants, and Pangaea's interior was extremely dry.

Dinosaurs first appeared around 230 Ma as small, two-legged predators such as Eoraptor and Herrerasaurus. Pterosaurs took to the skies around 228 Ma, and crocodilians diversified. Around 225 Ma the **first mammals** appeared: tiny, shrew-like, nocturnal insect-eaters.

Synapsids declined, except for the line leading to mammals, while the **archosaurs** — dinosaurs, pterosaurs, and crocodilians — rose to dominance, setting the stage for the Age of Dinosaurs.

Part 6 — Age of Giant Dinosaurs

Jurassic, 201–145 Ma. Gymnosperms reached their peak, with vast conifer forests over the continents and ferns in the understory. Carbon dioxide was 3–4 times today's level, keeping the climate warm and the vegetation lush.

Dinosaurs dominated. **Sauropods** such as Brachiosaurus and Diplodocus were the largest land animals that have ever lived. **Theropods** such as Allosaurus, about 9 m long, were the top predators, and Stegosaurus carried its famous back plates. Mammals stayed small and nocturnal.

Around 150 Ma **Archaeopteryx** appeared: the first known bird, with feathers and wings but also teeth, clawed fingers, and a bony tail. It marks the transition from theropod dinosaurs to birds.

Part 7 — The Flowering Plant Revolution

Cretaceous, 145–66 Ma. Flowering plants, or **angiosperms**, appeared around 140–130 Ma and diversified explosively; by the Late Cretaceous they dominated most environments. Magnolias are among the earliest, and have changed little since.

Their success came from **co-evolution with insects**. Flowers attracted pollinators such as bees, butterflies, and moths, and plants and pollinators diversified together. Fruits recruited animals to carry seeds away.

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Dinosaurs reached their peak diversity, including T. rex and Triceratops, while mammals and birds diversified. The **End-Cretaceous extinction** (66 Ma) was caused by the 10 km **Chicxulub asteroid** impact together with the Deccan volcanic eruptions. It ended the non-avian dinosaurs and the pterosaurs and killed about 75% of species, opening the door for mammals.

Part 8 — Age of Mammals & Humans

Cenozoic, 66 Ma to the present. Within about 10 million years of the dinosaurs' end, **mammals** had filled every major way of life. Primates appeared around 55 Ma and whales around 50 Ma, followed by horses, elephants, bats, and carnivores.

As the climate cooled, **grasslands** spread worldwide from around 23 Ma — savannas, prairies, and steppes — and grasses co-evolved with the grazing mammals that ate them.

Around 6–7 Ma the **hominin** line split from the other apes in Africa, and **Homo sapiens** evolved in Africa about 300,000 years ago. The ice ages of the last 2.6 million years brought mammoths, giant sloths, and saber-toothed cats, which thrived until humans arrived. Agriculture began about 11,700 years ago.

Part 9 — The Atmospheric Foundation

Nothing on the earlier pages could have happened without oxygen. **Cyanobacteria** began producing it by photosynthesis billions of years ago, and in the **Great Oxidation Event** (2.4 Ga) it began to build up in the atmosphere. From that oxygen, an **ozone layer** formed and began blocking the Sun's deadly ultraviolet (UV) light.

Without the ozone shield, UV light at ground level would be lethal to complex life. By about 470 Ma oxygen and ozone were sufficient for life to colonize the land, making them the prerequisite for all life on land.

Other forces shaped what came next. **Volcanic eruptions** released nutrients and carbon dioxide but also caused extinctions; **continental drift** assembled and broke up Pangaea, changing the climate; **asteroid impacts** cleared space for new groups; and ice ages and greenhouse periods forced life to adapt.

Part 10 — The Big Picture

Four innovations opened the land. **Vascular tissue** (430 Ma) let plants grow tall. **Seeds** protected plant embryos and freed plant reproduction from water. The **amniotic egg** (325 Ma) did the same for animals. **Flowers** (140 Ma), through co-evolution with pollinators, made angiosperms the most diverse plants on Earth.

Geologic forces drove the story too. Supercontinents formed dry interiors and, as they broke apart, created new and varied habitats. Mass extinctions — the End-Permian (252 Ma), which killed 96% of species, and the End-Cretaceous (66 Ma), which ended the dinosaurs — swept the stage clear, and each crisis created an opportunity for the survivors.

From microscopic crusts on bare rock to 50 m trees to human beings, the colonization of land took 500 million years, and it all started with oxygen.

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STUDY QUESTIONS

Questions with Model Answers

Question numbering matches the student study guide exactly. Model answers are written at the level expected of a strong student response, not as the only acceptable wording.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — The Pioneers

1. What were the earliest forms of life on land, and what did they do to the bare rock?

ANSWER Microscopic life: cyanobacteria and lichens, which formed thin crusts on bare rock. Lichens slowly weathered the rock into the first soil.

2. What is vascular tissue, and which plant was the first to have it?

ANSWER Internal plumbing (xylem and phloem) that carries water and food through a plant, letting it grow tall. Cooksonia, about 430 Ma, was the first vascular plant, though only 2–6 cm tall.

3. Name the first animals to venture onto land. What did they eat, and what kind of environment did they need?

ANSWER Millipedes (and, according to the exercise, scorpions), around 428 Ma. They fed on decaying plant material and each other, and needed moist environments.

Part 2 — The First Forests

1. What was Archaeopteris, and how tall did it grow?

ANSWER One of Earth's first true trees, forming the first forests around 385 Ma. It grew up to 30 m tall.

2. Why were Archaeopteris's deep roots so important?

ANSWER Roots more than 1 m deep broke up rock and created true soil, speeding up the weathering of rock.

3. What was Tiktaalik, and why is it called a “fishapod”?

ANSWER A 375-million-year-old transitional animal: a fish with limb-like fins. It shows part of the change from fish to four-limbed animals (tetrapods) — the vertebrate move from water to land.

Part 3 — The Coal Forests

1. How high did atmospheric oxygen reach in the Carboniferous, and how does that compare with today?

ANSWER About 35%, compared with 21% today.

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2. Give two examples of giant Carboniferous animals, and explain how the high oxygen level made their size possible.

ANSWER Any two of: the dragonfly Meganeura (70 cm wingspan); the millipede Arthropleura (2.5 m long). Insects and other arthropods take in oxygen by diffusion, which limits their size. More oxygen in the air let diffusion supply a larger body.

3. What is an amniotic egg, and why was it such an important innovation?

ANSWER An egg with its own water supply and a protective shell, which can be laid on dry land. It freed vertebrates from needing water to reproduce, around 325 Ma.

4. Where did today's coal come from?

ANSWER From the Carboniferous swamp forests. They were so productive that their remains were buried before they could decay, and were turned into coal seams.

Part 4 — Supercontinent Pangaea

1. Which plants dominated the Permian, and why did seed plants do better than spore plants?

ANSWER Gymnosperms (conifers, cycads, ginkgos) and seed ferns such as Glossopteris. Pangaea's interior was dry, and seeds do better than spores in arid environments.

2. What were gorgonopsids?

ANSWER Saber-toothed synapsid predators up to 3.5 m long, the top predators of the Permian.

3. What caused the End-Permian extinction, and how did it kill so many species?

ANSWER Massive volcanic eruptions in the Siberian Traps. They released carbon dioxide, causing global warming, ocean acidification, and oxygen-starved oceans. About 96% of species died out, 252 Ma.

Part 5 — Recovery & Revolution

1. When did the first dinosaurs appear, and what were they like?

ANSWER Around 230 Ma, in the Triassic. They were small, two-legged predators, such as Eoraptor and Herrerasaurus.

2. When did the first mammals appear, and what were they like?

ANSWER Around 225 Ma. They were tiny, shrew-like, nocturnal insect-eaters.

3. After the End-Permian extinction, which group declined and which group rose to dominance?

ANSWER Synapsids declined (except for the line leading to mammals). The archosaurs — dinosaurs, pterosaurs, and crocodilians — rose to dominance.

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Part 6 — Age of Giant Dinosaurs

1. What were sauropods? Name two.

ANSWER Giant, long-necked, plant-eating dinosaurs, the largest land animals ever. Examples: Brachiosaurus, Diplodocus.

2. What was Archaeopteryx, and when did it live?

ANSWER The first known bird, about 150 Ma. With feathers and wings but also teeth, clawed fingers, and a bony tail, it is transitional between theropod dinosaurs and birds.

3. Describe the Jurassic atmosphere and climate, and their effect on plant life.

ANSWER Carbon dioxide was 3–4 times today's level, making the climate warm and the vegetation lush, with vast conifer forests covering the continents.

Part 7 — The Flowering Plant Revolution

1. When did flowering plants appear, and what is their scientific name?

ANSWER Around 140–130 Ma, in the Cretaceous. They are the angiosperms.

2. Explain how flowering plants and insects co-evolved, and what fruits added.

ANSWER Flowers attracted insect pollinators such as bees, butterflies, and moths. Each group adapted to the other, so both diversified rapidly together. Fruits got animals to eat them and spread the seeds.

3. What caused the End-Cretaceous extinction, and what did it make possible?

ANSWER The impact of the Chicxulub asteroid, about 10 km wide, together with the Deccan volcanic eruptions, 66 Ma. It killed the non-avian dinosaurs, the pterosaurs, and about 75% of species, opening the way for mammals to dominate.

4. Why do scientists say the non-avian dinosaurs died out, rather than all the dinosaurs?

ANSWER Because birds are descended from theropod dinosaurs — they are the avian dinosaurs, and they survived.

Part 8 — Age of Mammals & Humans

1. How quickly did mammals fill the roles left open by the dinosaurs? Name three groups that appeared.

ANSWER Within about 10 million years. Any three of: primates (~55 Ma), whales (~50 Ma), horses, elephants, bats, carnivores.

2. When and why did grasslands spread, and how did they affect mammals?

ANSWER From around 23 Ma, driven by a cooling climate. Grasses co-evolved with the grazing mammals that ate them.

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3. When did the hominin line split from the other apes, and when and where did Homo sapiens evolve?

ANSWER About 6–7 Ma, in Africa. Homo sapiens evolved about 300,000 years ago, also in Africa.

Part 9 — The Atmospheric Foundation

1. What organisms first produced Earth's oxygen, and when did it begin to build up in the atmosphere?

ANSWER Cyanobacteria, by photosynthesis. Oxygen began to build up in the Great Oxidation Event, about 2.4 Ga.

2. Why was the ozone layer necessary before complex life could live on land?

ANSWER It blocks the Sun's ultraviolet light. Without it, UV at ground level would be lethal to complex life.

3. Besides oxygen, name two geologic forces that drove the evolution of life on land, and say how.

ANSWER Any two of: volcanic eruptions (released nutrients and carbon dioxide, but also caused extinctions); continental drift (assembling and breaking up Pangaea changed the climate); asteroid impacts (cleared space for new groups); climate cycles (ice ages and greenhouse periods forced adaptation).

Part 10 — The Big Picture

1. Name the four major innovations on the Big Picture page, and say what each one made possible.

ANSWER Vascular tissue let plants grow tall; seeds freed plant reproduction from water; the amniotic egg freed animal reproduction from water; flowers, through pollinators, drove the diversification of angiosperms.

2. How did the forming and breaking up of supercontinents affect evolution?

ANSWER A supercontinent has a dry interior, which favored drought-tolerant life such as seed plants. Its breakup created new and varied habitats for life to diversify into.

3. “Each crisis created opportunity.” Explain this using one of the two great mass extinctions.

ANSWER End-Cretaceous: the asteroid ended the non-avian dinosaurs, and mammals then filled their roles. Or End-Permian: the synapsids that had ruled the Permian declined, and the archosaurs, including the dinosaurs, rose to take their place.

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ANSWER KEY

The 10 Graded Slide-Quiz Questions

The questions students answer inside the exercise, in page order.

Page	Q	Question	Keyed answer
1. Pioneers	1	First vascular plant on land?	B) Cooksonia
2. First Forests	2	What was revolutionary about Archaeopteris?	B) Deep roots (>1 m) that created true soil
3. Coal Forests	3	Oxygen level that enabled giant insects?	C) 35%
4. Pangaea	4	Cause of the End-Permian extinction?	C) Siberian Traps eruptions
5. Triassic	5	When did the first dinosaurs appear?	B) ~230 Ma
6. Jurassic	6	Transition between dinosaurs and birds?	B) Archaeopteryx
7. Cretaceous	7	What was revolutionary about angiosperms?	C) Co-evolved with insect pollinators
8. Cenozoic	8	When did Homo sapiens evolve?	C) ~300,000 years ago
9. Atmosphere	9	Primary factor that enabled life to colonize land?	B) Oxygen and the ozone layer
10. Big Picture	10	Innovation that freed animals from water reproduction?	B) The amniotic egg

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TEACHER NOTES

Corrections and Cautions

Points in the exercise that need a correction, a qualification, or a prepared answer for the question a sharp student will ask.

C1 **Archaeopteris did not have seeds, and seeds are not as old as 385 Ma.**

Page 2 lists "early seed plants (progymnosperms)." Progymnosperms, Archaeopteris among them, reproduced by spores; they were the ancestors of seed plants, not seed plants themselves. The earliest known seeds date to about 365 Ma, late in the Devonian. The Big Picture page dates seeds to 385 Ma, which is really the date of the first forests. The graded question is unaffected ("they reproduced with seeds" is correctly a wrong answer), and the study guide avoids the date.

C2 **No flying insects in the Devonian.**

Page 2 says Devonian insects began developing flight. The oldest winged insects are Carboniferous, around 325 Ma, and Devonian insect fossils are very rare. The study guide leaves this out.

C3 **"Earth's first tree."**

Archaeopteris is the first tree with modern features — true wood, leaves, and deep roots — but Wattieza (Eospermatopteris) formed forests at Gilboa, New York, at about the same time or slightly earlier. The study guide says "one of Earth's first true trees."

C4 **Early tetrapods and early scorpions were mostly in the water.**

Acanthostega was largely aquatic, and Ichthyostega could at most move awkwardly on land, so "first tetrapods walk on land" overstates it. Likewise, Silurian scorpions were probably still aquatic; the 428 Ma land pioneer is the millipede Pneumodesmus. Lichens before about 410 Ma are uncertain in the fossil record, while microbial crusts on land are far older than 500 Ma.

C5 **"96% of all species" is a marine figure, and an upper one.**

The 96% End-Permian figure is an older estimate for marine species; recent estimates run around 81% for marine species and about 70% for land vertebrate species. It is still the worst extinction known. On the same page, Scutosaurus, listed under "Synapsids Rule," was a pareiasaur, a reptile rather than a synapsid; Moschops is correctly a synapsid.

C6 **Jurassic sauropod sizes are inflated.**

"30+ m long, 80+ tons" overstates Brachiosaurus (about 22 m, perhaps 30–50 tons) and Diplodocus (about 25 m, about 15 tons). The 70–80-ton giants, such as Argentinosaurus, were Cretaceous titanosaurs. "Largest land animals ever" is correct for the sauropods as a group.

C7 **Cretaceous grasses, oaks, and maples.**

Grasses existed by the Late Cretaceous but were minor; grasslands are a Cenozoic story, as page 8 says. Oaks and maples are known mainly from the Cenozoic. Magnolia is a good Cretaceous example.

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C8 **Oxygen did not “stabilize around 21%” from 299 Ma.**

Estimates show oxygen falling sharply after the Carboniferous–Permian peak, perhaps to 15% or less around the Permian–Triassic boundary, and fluctuating since. The 3.5 Ga start of oxygen-producing photosynthesis on page 9 is also contested; firm evidence begins nearer 3.0–2.7 Ga. All ancient oxygen figures, including the 35% peak, are model estimates with wide error bars.

C9 **Page wording.**

The outro page heading reads “What You Learned in This Module”; the guides call it an exercise.

TEACHER NOTES

Teaching the Exercise

Put the timeline on a human scale.

Squeeze the 500 million years into one calendar year. The first forests arrive in late March, the first dinosaurs in mid-July, the asteroid hits in mid-November, Homo sapiens appears about five hours before midnight on New Year's Eve, and agriculture begins about 12 minutes before midnight.

Tie it back to astronomy.

Oxygen and ozone are exactly what astronomers hope to detect in the atmospheres of exoplanets. This exercise shows why: on Earth, they are the signature of photosynthesis, and the shield that made land life possible. It also took about two billion years after the Great Oxidation Event for life to reach the land — a useful caution about how long a habitable planet may need.

Make the Chicxulub link explicit.

The End-Cretaceous extinction is an astronomical event with biological consequences. It connects naturally to near-Earth asteroids, impact frequency, and planetary defense.

Ask why the giant insects are gone.

Students often assume big bugs went extinct for no reason. The diffusion argument — an insect's oxygen supply comes in through tubes, so size depends on how much oxygen is in the air — is a good example of physics setting a limit on biology.

Use the pattern of the two extinctions.

Both great extinctions involve huge volcanic outpourings (Siberian Traps, Deccan Traps), and both were followed by a new group taking over. Ask students to compare them before they reach the Big Picture page.

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